HEBEI HANDAN DAMINGCHANG BIOGAS RECOVERY AND UTILIZATION PROJECT

Document Prepared by Climate Bridge (Shanghai) Ltd.

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Project Title	Hebei Handan Damingchang Biogas Recovery and Utilization Project
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Hebei Handan Damingchang Biogas Recovery and Utilization Project (hereinafter referred to as the Project) locates in Cao Ren Village, Dangtou Township, Daming County, Handan City, Hebei Province, P.R China. The project is owned and implemented by Daming Jiujin Energy Co., Ltd.

The project is to build a centralized anaerobic animal manure treatment system which collects manure waste from dairy cows from Daming farms in Daming County. The project uses animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. The recovered biogas will be used for electricity generation at the project site. The electricity generated by the project will be delivered to North China Power Grid (NCPG). The residual waste from the CSTR digesters will be used to produce organic fertilizer at the project site.

The project started construction on 06-February-2022 and put into operation on 26-August-2022.

Prior to the implementation of the project, the animal manure waste was left to decay in anaerobic manure management system (open lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility. Equivalent amount of electricity was supplied from the NCPG connected fossil fuel fired power plants.

The project is expected to avoid GHG emission of methane through recovery and destruction of biogas. In addition, the electricity generated from the biogas will be delivered to NCPG to replace equivalent electricity generated by the fossil fuel fired power plants connected to the grid. The project is expected to achieve an annual emission reduction of 28,723 tCO_2e and a total emission reduction of 201,061 tCO_2e during the first 7-year crediting period.

1.2 Sectoral Scope and Project Type

Sectoral Scope 1: Energy industries (renewable - / non-renewable sources)

Sectoral Scope 13: Waste handling and disposal.

The project is not a grouped project.

1.3 Project Eligibility

The scope of the VCS Program includes:

1) The six Kyoto Protocol greenhouse gases: The project is expected to avoid two greenhouse gases: 1) Methane (CH_4) emissions from the anaerobic animal manure management system in

the baseline scenario, which will be captured and destroyed in the project scenario; 2) CO_2 emissions from the production of the equivalent amount of electricity replaced by the Project that would otherwise be generated by the fossil fuel fired power plants of NCPG. Thus, the project applicable to this scope.

2) Ozone-depleting substances: Not Applicable.

3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process: Not Applicable.

4) Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval: The applied methodology AMS-III.D (Version 21.0) and AMS-I.D (Version 18.0) of the project are methodologies approved under CDM Program, which is a VCS approved GHG program.

5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: Not Applicable.

Furthermore, the project does not belong to the project activities excluded in Table 1 of VCS Standard 4.3.

Thus, the project is eligible under the scope of VCS program.

1.4 Project Design

The project has been designed to be a single installation of an activity, not a grouped project.

Eligibility Criteria

The project is not a grouped project.

1.5 Project Proponent

Organization name	Daming Jiujin Energy Co., Ltd
Contact person	ZHANG Kai
Title	General Manager
Address	Cao Ren Village, Dangtou Township, Daming County, Handan City, Hebei Province, P.R China
Telephone	+86 21 23019950
Email	3542346576@qq.com

1.6 Other Entities Involved in the Project

Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	VCUs Buyer
Contact person	Gao Zhiwen
Title	General Manager
Address	Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, China 200120
Telephone	+86-21 62462036
Email	gao.zhiwen@climatebridge.com

1.7 Ownership

The project owner of the project is Daming Jiujin Energy Co., Ltd. The approval of project, Environmental Impact Assessment (EIA) and the business license of the project owner are evidence for legislative right.

1.8 Project Start Date

26-August-2022 (The date when the generator was first connected to the grid for power generation)

1.9 Project Crediting Period

This project adopts a 7-year renewable crediting period, from 26-August-2022 to 25-August-2029 (both days included).

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

☐ <20,000 tCO_2e/year

☒ 20,000 – 100,000 tCO_2e/year

☐ 100,001 – 1,000,000 tCO_2e/year

☐ >1,000,000 tCO_2e/year

Project Scale	
Project	✓
Large project	

Year	Estimated GHG emission reductions or removals (tCO_2e)
26-August-2022~31-December-2022	10,073[1]
01-January-2023~31-December -2023	28,723
01-January -2024~31-December -2024	28,723
01-January -2025~31-December -2025	28,723
01-January -2026~31-December -2026	28,723
01-January -2027~31-December -2027	28,723
01-January -2028~31-December -2028	28,723
01-January -2029~25-August -2029	18,650[2]
Total estimated ERs	201,061
Total number of crediting years	7
Average annual ERs	28,723

1.11 Description of the Project Activity

The project involves construction of a centralized animal manure management system which collects manure waste from dairy cow from Daming farms in Daming county. The project uses animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. The anaerobic digesters are expected to produce 8,760,000 m^3 biogas per year. The recovered biogas will be used for electricity generation at the biogas power plant. Waste heat recovery boiler will also be installed at the biogas power plant to recover and supply waste heat to the auxiliary equipment of the project. The emission reductions from waste heat recovery will not be accounted for. The installed capacity of the biogas power plant will be 2.33MW. The biogas power plant of the project is expected to supply 16,510MWh of

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[1] It covered 128 days from 26-August-2022~31-December-2022, thus the estimated emission reduction during this period is 10,073 tCO_2e = 28,723 tCO_2e x 128/365.

[2] It covered 237 days from 01-January-2029~25-August-2029, thus the estimated emission reduction during this period is 18,650 tCO_2e = 28,723 tCO_2e x 237/365.

electricity to the North China Power Grid annually. The project activity is expected to have a 20-year operational lifetime.

The project is expected to avoid GHG emission of methane through recovery and destruction of biogas. In addition, the electricity generated from the biogas will be delivered to NCPG to replace equivalent electricity generated by the fossil fuel fired power plants connected to the grid.

The key system involved in the project are as follows:

Anaerobic animal manure management system:

The Completely Stirred Tank Reactor (CSTR) type anaerobic digesters are to be applied in the project activity. Three CSTR digesters with effective volume of 6468.33m^3 each will be installed. The project is expected to produce 8,760,000 m^3 biogas per year.

Biogas pre-treatment system

Before electricity generation, combustion or flaring, the biogas will be pre-treated to remove impurities and moisture etc., to prevent the corrosion of the project facility. In addition, biogas should be continuously in a stable condition before it flows into gas engine or flare. The pre-treatment consists of pre-filtration, dehumidification, dewatering, cooling and fine filtration.

Organic fertilizer production plant

The digestate from the anaerobic digesters, both solid and liquid digestate, will be used as material to produce organic fertilizer at the onsite organic fertilizer production plant. Hence proper conditions and procedures (not resulting in methane emissions) for digestate is ensured.

Biogas power generation system

The biogas electricity generation plant will be installed at the project site. The total installed capacity will be 2.33MW (1.165MW×2). The biogas power plant of the project is expected to supply 16,510 MWh of electricity to the North China Power Grid annually. Waste heat recovery boiler will also be installed at the biogas power plant to recover and supply waste heat to the auxiliary equipment of the project.

1.12 Project Location

The Project locates in Cao Ren Village, Dangtou Township, Daming County, Handan City, Hebei Province, P.R China.

The geographical coordinates for the project site are east longitude 115°17’0.405’’ and north latitude 36°11’20.403’’.

The geographic location of the project is shown in Figure 1.

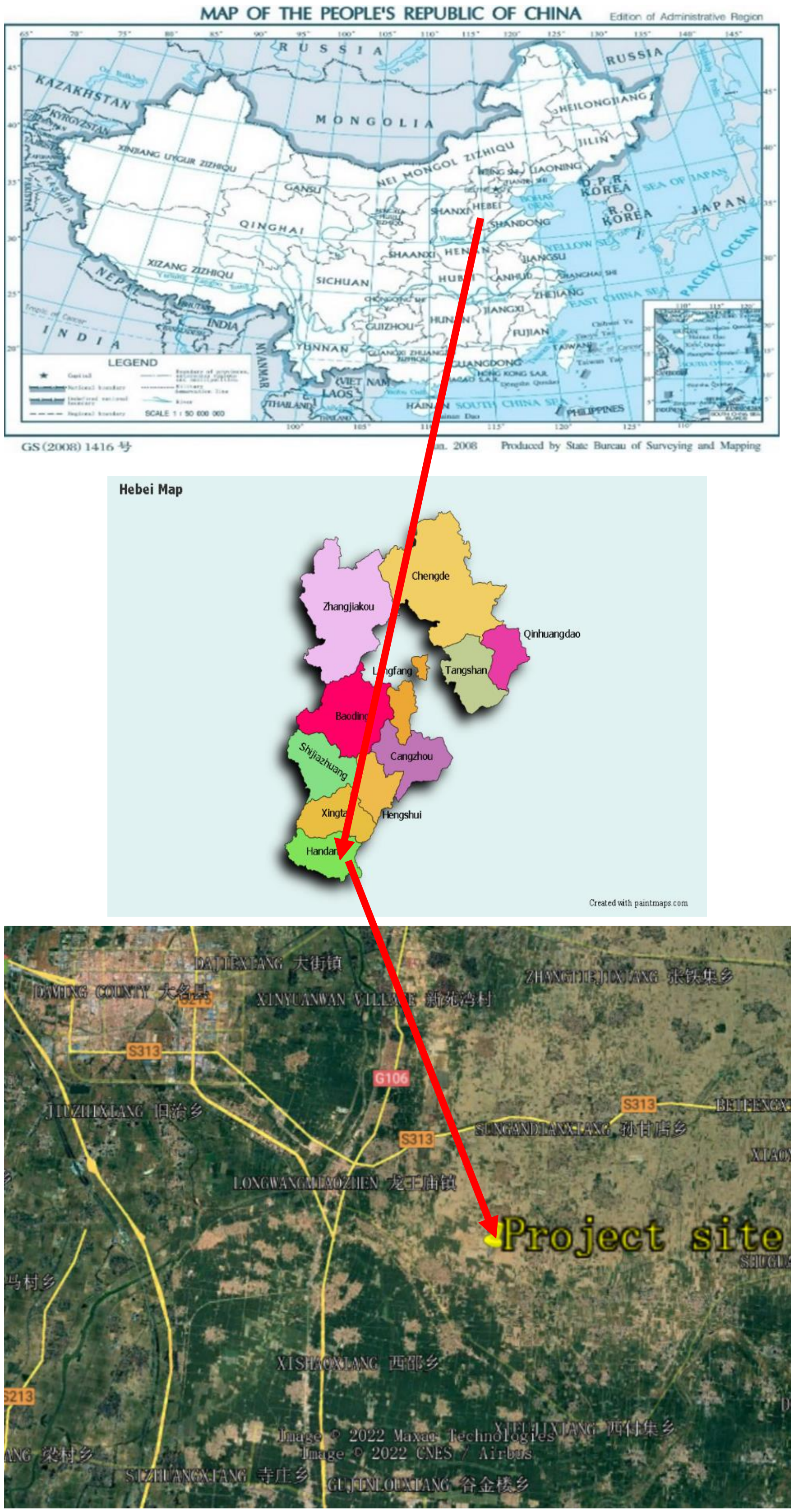


Figure 1: Location of the project

1.13 Conditions Prior to Project Initiation

The conditions existing prior to project initiation:

1) The animal manure waste was left to decay in anaerobic manure management system (lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility.

2) Equivalent amount of electricity was supplied from the NCPG connected fossil fuel fired power plants.

The conditions existing prior to project initiation is also the baseline scenario of the project.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Chinese relevant laws and regulations. Mainly include:

No.	Laws and Regulations	Compliance of the project
1	Law of the People's Republic of China on the Prevention and Control of Solid Waste Pollution;	*Clause 65 of the Law stipulate "Those engaged in large-scale livestock and poultry breeding shall promptly collect, store, utilize or dispose if solid waste such as livestock and poultry manure generated in the breeding process to avoid environmental pollution"*. The project is to build a centralized anaerobic animal manure treatment system, which fully complies with the law.
2	Regulations on prevention and control of pollution from large scale livestock and poultry breeding	*The Regulation requires effective measures such as scientific breeding methods and waste treatment techniques should be adopted to reduce the amount of waste generated from livestock and poultry breeding and the amount of pollutants emissions to the environment.* The project adopts CSTR to treat animal manures from livestock farms, which fully complies with the Regulation.

The project has obtained the project approval and EIA approval from local government authorities: Daming County Administrative Examination and Approval Bureau. The two approvals well demonstrate that local government permits the construction of the project. Consequently, the project is compliance with laws, status and other regulatory frameworks.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has neither been registered, nor is seeking registration under any other GHG programs. The project is seeking registration only under VCS program.

1.15.2 Projects Rejected by Other GHG Programs

The project has never been seeking registration under any other GHG program; hence the project has never been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project reduces GHG emissions from activities that are not included in an emissions trading program or any other mechanism that includes GHG allowance trading.

The project proponent is not part of any emission trading program. The net GHG emission reductions from the project will not be used for compliance with emission trading programs or to meet binding limits on GHG emissions. The project activity has not participated under any other GHG programs.

China has a national emissions trading scheme only cover the high-emission industries, such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO_2e/year[3]. And the project activity is not included the mandatory emission control scheme and there is no emission cap enforced for the project owner according to the enforced company list[4] in public information. Hence, it is confirmed that the emission reductions will not be double counted.

1.16.2 Other Forms of Environmental Credit

The project hasn't sought or received another form of environmental credits.

1.17 Sustainable Development Contributions

A summary description of project activities that result in sustainable development (SD) contributions:

The project uses animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. The recovered biogas will be used for electricity generation at the project site. The electricity generated by the project will be delivered to North China Power Grid (NCPG). The residual waste from the CSTR digesters will be used to

[3] http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330_826728.html

[4] http://mee.gov.cn/xxgk2018/xxgk/xxgk03/202012/W020201230736907682380.pdf

produce organic fertilizer at the project site. The project has positive impact to climate and also provide jobs.

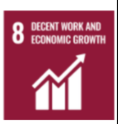
The Project will contribute to sustainable development in the following ways:

- The project activity will enhance the quality of the water, decrease odour nuisance, improve the working environment of the workers of the livestock farm, and minimize health risks of local residents. Thus, the project will achieve SDG 6 "Ensure availability and sustainable management of water and sanitation for all".
- The project will achieve a GHG emission reduction during the crediting period. Thus, the project will achieve SDG 13"Take urgent action to combat climate change and its impacts".
- This project will increase income of local residences and accelerate economy development in rural areas. During the crediting period, direct and indirect employment opportunities will be generated. Thus, the project will achieve SDG 8 "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all".

The Project will contribute to sustainable development, please check the following table:

SDG	Indicators	Chinese Sustainable Development Progress	Project activity contribution
13 CLIMATE ACTION	SDG 13 "Take urgent action to combat climate change and its impacts"[5].	In 2020, China's energy consumption per unit of GDP was reduced by 24.4% compared with 2012; carbon dioxide emissions per unit of GDP was reduced by 18.8% compared with 2015 and 48.4% compared with 2005, all of which have already fulfilled China's commitment to the international community in 2020 ahead of schedule.	The project uses CSTR to treat animal manure, collect and destroy the generated biogas to avoid methane emissions. This contributes to achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas"
6 CLEAN WATER AND SANITATION	SDG 6 "Ensure availability and sustainable management of	The Chinese government promotes water control, and promotes the conservation, protection and management of water resources. The	The project activity will enhance the quality of the water, decrease odour nuisance, improve the working environment of the workers of

[5] https://sdgs.un.org/goals/goal13

	water and sanitation for all".[6]	urban and rural water supply and environmental sanitation conditions have been significantly improved, and the people feel happy. It is of great significance to promote the comprehensive green transformation of economic and social development.	the livestock farm, and minimize health risks of local residents. This contributes to one of China's actions for promoting sustainable developing: "Continue to implement water pollution prevention and control. Coordinate the development of high quality protection and high levels of development and continuously improve the ecological environment of the basin.
	SDG 8 "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all"[7].	China continuously improves the quality and efficiency of development. In-depth implementation of the innovation-driven development strategy, the rapid development of small and medium-sized enterprises. Adhering to the policy of giving priority to employment, the unemployment rate has remained at a low level. By coordinating epidemic prevention and control and economic and social development, it has become the only major economy to achieve positive growth in 2020 and has made positive contributions to the recovery of the global economy.	The project activity will increase employment opportunities for the operation of the project activity. This contributes to one of China's actions for promoting sustainable developing: "by 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value".

1.18 Additional Information Relevant to the Project

[6] https://sdgs.un.org/goals/goal6

[7] https://sdgs.un.org/goals/goal8

Leakage Management

As per “Project and leakage emissions from anaerobic digesters” (version 02.0), the leakage emissions associated with the anaerobic digester depend on how the digestate is managed. The leakage emissions include emissions associated with storage of digestate and composting of the digestate.

Digestate from the anaerobic digesters of the project will be sent to aerobic composting as soon as digestate is removed from the digesters and after composting, it will be returned to nearby farm as fertilizer. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve anaerobic composting of digestate. Therefore, leakage emissions of the project associated with the anaerobic digester is not accounted for.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable.

2 SAFEGUARDS

2.1 No Net Harm

The Environmental Impact Assessment (EIA) of the project has been conducted, and EIA report has been approved by Daming County Administrative Examination on 09-September-2021 (Approval No. "Dashenhuanping[2021] No.18"). The EIA report has assessed every possible aspect of environmental impact of the project and recommended corresponding measures, where applicable. No net harm has been detected. The project will also contribute to the sustainable development of local community as described in section 1.17 above.

In conclusion, the project has no negative impacts on local environment. No net harm on local environment and social community has been detected for the project.

2.2 Local Stakeholder Consultation

The Project owner collected comments by local stakeholders on the project activity. Survey questionnaires were distributed to relevant personnel of the livestock farms, local villagers and government officials by the Project owner on 08-July-2021. The survey questionnaire was designed to assess the project impacts on the local environment and social economic development. The structure of the survey respondents is listed in Table 1-1 below.

Table 1-1: Structure of stakeholder survey

Item	Distribution	Quantity	Percentage
Amount of stakeholders surveyed	Male	18	60%
	Female	12	40%
Age	<25	6	20%
	25-55	18	60%
	>55	6	20%
Education	Junior high school or below	10	33%
	Senior high school	12	40%
	College or above	8	27%
Occupation	Worker	16	53%
	Peasant	9	30%
	Management personnel	2	7%
	Civil servant	2	7%

	Unspecified	1	3%

30 questionnaires were distributed to local stakeholders, and all questionnaires have been recollected. Comments from these questionnaires are summarized in Table 2-1 below:

No.	Questions	Attitude or Opinion	Amount	Percentage
1	Do you know about the project activity?	Very much	21	70%
		Heard of	5	17%
		Nothing	4	13%
2	Do you think the project will improve the current situation of livestock farms?	Yes	27	90%
		No	0	0
		Don't know	3	10%
3	Do you think the project will improve the local employment situation?	Yes	20	67%
		No	0	0
		Don't know	10	33%
4	Do you think the project will improve the local social community?	Yes	19	63%
		No	0	0
		Don't know	11	37%
5	What is the most probable environmental impact do you think the project will cause after the construction finish? (multiple choice)	None	23	77%
		Air pollution	0	0%
		Water pollution	0	0
		Noise pollution	0	0%
		Harm to indigenous animals and plants	0	0%
		Don't know	7	23%
6	What is your attitude to the project activity?	Support	26	87%
		Against	0	0
		Indifferent	4	13%

In general, local stakeholders are supportive of the project construction. The survey shows that a majority of local stakeholders think the Project will help improve the life of local people without much adverse environmental impact. The survey shows that almost all of the stakeholders are supportive to the proposed project, believing that the Project will provide more employment opportunities and improve the local air quality by reducing the diffusion of odorous gases. Therefore, the implementation of the Project is regarded as beneficial by most of the local stakeholders. In order to understand the opinions of various stakeholders during the project operation period, the project owner will set up an on-going communication mechanism to regularly holds stakeholder meetings every year to issue questionnaires and to answer the questions from various stakeholders.

2.3 Environmental Impact

As per the EIA report, the environmental impacts of the project in construction period and operation period are summarized as follows.

1. <u>Construction Phase</u>

1.1 Air pollution

Construction dust and road dust during the construction has a certain effect to the surrounding areas of construction site, these adverse effects are accidental, temporary and partial, as long as to take effective measures and strengthen management, the scope of its influence is generally limited to the surrounding area of the construction site and will disappear along with the end of construction.

1.2 Wastewater

Wastewater during the construction period mainly includes construction wastewater and domestic sewage. The following measures are taken to prevent and control the pollution of construction wastewater: construct temporary diversion ditches on the construction site; Set up a sedimentation tank, and reuse the washing water for construction machinery as much as possible after simple treatment of equipment and vehicles;

Domestic sewage can be collected and processed by setting up mobile toilets, and regularly handed over to the sanitation department to remove garbage and excrement, which can effectively prevent the sewage generated by construction workers from polluting the water environment.

1.3 Noise

The noise generated by the project construction has a slight impact on the surrounding sensitive points, and its pollution impact is localized and short-term. After adopting a reasonable construction organization method, its impact on the surrounding area can be reduced to acceptable range.

1.4 Solid waste

During the construction period, a certain amount of construction waste and domestic waste from construction workers will be generated, of which part of the construction waste will be recycled, and the unusable waste will be handed over to the sanitation department for disposal. After the above measures are taken, the solid waste of the project will not cause pollution to the surrounding environment.

2. <u>Operation Phase</u>

2.1 Air pollution

The air pollution during the operation mainly includes the exhaust gas of the biogas generator and the flare, and dust from organic fertilizer plant. The exhaust gas is to be treated through 2 stage purification equipment before being discharged through exhaust cylinders with 20m high. The dust from organic fertilizer plant is to be filtered by dust collector and reused in the organic fertilizer plant. The NO_x, sulfur dioxide, H_2S, NH_3 and particulate matter in all the exhaust gas by the project meets Grade 2 emission standard of "Integrated Emission Standard of Air Pollutants " (GB16297).

2.2 Wastewater

The wastewater during operation is mainly domestic sewage and biogas slurry from anaerobic digesters. The domestic sewage of the project is pretreated in a septic tank and reused in anaerobic digesters. Most biogas slurry will also be recycled in anaerobic digesters, surplus biogas slurry will be used as material for organic fertilizer production. No wastewater will be discharged to the environment.

2.3 Noise

The noise of this project comes from noise from various mechanical operation and vibration. All production equipment is placed in the workshop or steel container, and measures are taken to reduce vibration and noise. The operating noise is effectively attenuated after being blocked by the solid wall.

2.4 Solid waste

The solid waste of the project is mainly domestic garbage, filtered dust. The domestic garbage and filtered dust are collected uniformly and then sent to landfill site for landfill treatment.

Hazardous waste: The hazardous waste, e.g. mineral oil, generated by the project mechanical facilities shall be properly collected and stored in the hazardous waste temporary deposit, and shall be submitted to the relevant qualified entities for centralized treatment regularly.

In conclusion, the environmental impact during the project construction will be temporary and not significant, and the environmental impact during the project operation will be minor. The project activity can reduce greenhouse gas emissions and environmental pollution caused by methane release and coal-fired power generation. The Project owner takes appropriate measures to minimize adverse environmental impacts.

2.4 Public Comments

This project will open for public comment on the verra website.

2.5 AFOLU-Specific Safeguards

The project is not a AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The methodologies applied to the project are small scale CDM methodologies:

AMS-III.D Methane recovery in animal manure management systems (version 21.0);

AMS-I.D Grid connected renewable electricity generation (version 18.0).

This methodology also refers to the latest approved version of the following tools:

Project and leakage emissions from anaerobic digesters (version 02.0);

Tool to calculate the emission factor for an electricity system (version 07.0);

For more detail information about the methodology and tools, please reference to the following link:

https://cdm.unfccc.int/methodologies/DB/H9DVSB24O7GEZQYLYNWUX23YS6G4RC

https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK

3.2 Applicability of Methodology

The project satisfies all the applicability criteria of the methodology AMS-III.D (Version 21.0) and AMS-I.D (version 18.0), of which the detailed description is listed in Table 3-1 below:

Table 3-1 Applicability of AMS-III.D

No.	Applicability conditions of the methodology	The project
1	This methodology covers project activities involving the replacement or modification of anaerobic animal manure management systems in livestock farms to achieve methane recovery and destruction by flaring/combustion or gainful use of the recovered methane. It also covers treatment of manure collected from several farms in a centralized plant	Applicable. The project is a centralized plant treats animal manure waste collected from Daming farms. The project replaces existing anaerobic animal manure management systems (open lagoon) in livestock farms to achieve methane recovery and destruction by flaring/combustion or gainful use of the recovered methane.
2	This methodology is only applicable under the following conditions:	Applicable.

	a) The livestock population in the farm is managed under confined conditions	All the livestock population in the farms within the project boundary is managed under confined conditions
	b) Manure or the streams obtained after treatment are not discharged into natural water resources (e.g. river or estuaries), otherwise "AMS-III.H Methane recovery in wastewater treatment" shall be applied	Applicable. As per the FSR, manure or the streams obtained after treatment are not discharged into natural water resources (e.g. river or estuaries)
	c) The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C	Applicable. The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C. The average temperature in project site is 13.5°C[8].
	d) In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m	Applicable. In the baseline scenario the retention time of manure waste in the anaerobic lagoons is greater than one month, and their depths are at least 1 m.
	e) No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario	Applicable. No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario.
3	The project activity shall satisfy the following conditions: (a) The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of "AMS-III.AO Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions and procedures (not resulting in methane emissions) must be ensured;	Applicable. The residual waste from the animal manure management system of the project will be used to produce organic fertilizer, which is handled aerobically and will not result in methane emissions.

[8] http://www.weather.com.cn/cityintro/101091001.shtml

	(b) Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared;	Applicable. Biogas produced by the project are used directly to produce electricity, the emergency flare ensure biogas would be destroyed while exigencies happened
	(c) The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when removed from the animal barns is larger than 20%, this time constraint will not apply.	Applicable. The storage time of the manure after removal from the animal barns, including transportation will not exceed 45 days before being fed into the anaerobic digester.
4	Projects that recover methane from landfills shall use “AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure shall use “AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology “AMS-III.AO Methane recovery through controlled anaerobic digestion".	Irrelevant. The project does not involve landfill methane recovery, wastewater treatment, composting animal manure, or co-digestion of animal manure and other organic matters.
5	Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the project activity, it should be taken into account accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component.	Applicable. Part of the recovered biogas is used to produce and deliver electricity to the power grid, therefore AMS.I.D is applied. Part of the biogas is used to power auxiliary equipment of the project activity, zero is used as its emission factor, and this part of biogas is not eligible as an SSC CDM Type I project component, which means no emission reductions will be claimed for this part of biogas utilization.
6	New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant	Applicable. The project is a Greenfield project. The emission reduction sourced from methane

	requirements in the "General guidelines for SSC CDM methodologies".	recovery is 28,723 tCO_2e/yr, which is lower than the threshold of 60,000 tCO_2e/yr; the installed capacity of the project is 2.33MW, which is lower than the threshold of 15MW. Therefore, the Project is in line with " General Guidelines to SSC CDM methodologies"
7	The requirements concerning demonstration of the remaining lifetime of the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies".	The project is a Greenfield project, does not involve the replaced equipment; therefore, this is irrelevant.
8	Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO_2 equivalent annually from all Type III components of the project activity.	Applicable. The emission reductions from the recovery and destruction of methane (viz. Type III components of the project) is 28,723 tCO_2e/yr, which is less than 60 kt CO_2 equivalent.

Table 3-2 Applicability of AMS-I.D

No.	Applicability conditions of the methodology	The project
1	This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	Applicable. The project will use biogas to generate electricity, and the electricity will be supplied to North China Power Grid.
2	Illustration of respective situations under which each of the methodology (i.e. "AMS-I.D.: Grid connected renewable electricity generation", "AMS-I.F.: Renewable electricity generation for captive use and mini-grid" and "AMS-I.A.: Electricity generation by the user) applies is included in the appendix.	The illustration of respective situations for AMS-I.D stated in the appendix is the same as applicability condition No. 1 above. No need for further justification.

3	This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	Applicable. The project is a greenfield plant.
4	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m2; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m2.	Irrelevant. The project is not a hydro power plant.
5	If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	Irrelevant. The project does not involve non-renewable components.
6	Combined heat and power (co-generation) systems are not eligible under this category.	According to the FSR, the only output for the project is electricity and the project does not involve co-generation system.

7	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct1 from the existing units.	Irrelevant. The project does not involve capacity addition.
8	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	Irrelevant. The project does not involve retrofit, rehabilitation or replacement.
9	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored.	Irrelevant. The project does not involve landfill gas, waste gas, wastewater treatment and agro-industries projects.
10	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	Irrelevant. The project does not involve biomass.

In addition, the project meets the applicability conditions of the applied tools applied in the PD as follows:

Table 3-3 Applicability of applied tools

Tool	Applicability	The project
Tool to calculate the emission factor for an electricity system (version 07.0)	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid	Applicable. The project utilizes biogas for power generation to supply electricity to NCPG that

	electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects)	dominated by fossil-fuel plants. Therefore the tool is applied to estimate the OM, BM and/or CM when calculating baseline emissions for the project.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option II a and option II b. If option II a is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	In the host country as off-grid power generation is not significant. Therefore, emission factor for the project electricity system is calculated only for the grid power plants. Thus, this applicability criterion is satisfied.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Applicable. The project electricity system is located in a non-Annex I country.

	Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	The project is not involved in biofuels.
Project and leakage emissions from anaerobic digesters (version 02.0)	The following sources of project emissions are accounted for in this tool: (a) CO_2 emissions from consumption of electricity associated with the operation of the anaerobic digester; (b) CO_2 emissions from consumption of fossil fuels associated with the operation of the anaerobic digester; (c) CH_4 emissions from the digester (emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester); and (d) CH_4 emissions from flaring of biogas.	Applicable. All sources of project emissions have been accounted.
	The following sources of leakage emissions are accounted for in this tool: (a) CH_4 and N_2O emission from composting of digestate; (b) CH_4 emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond.	Irrelevant. The project does not involve composting, anaerobic decay or anaerobic storage of digestate.
	Emission sources associated with N_2O emissions from physical leakages from the digester, transportation of feed material and	Applicable. As per the applied methodology, N_2O emissions are neglected because these are minor emission sources.

	digestate or any other on-site transportation, piped distribution of the biogas, aerobic treatment of liquid digestate and land application of the digestate are neglected because these are minor emission sources or because they are accounted in the methodologies referring to this tool.	

3.3 Project Boundary

According to the methodology AMS-III.D, the project boundary includes the physical, geographical site(s) of (a) The livestock; (b) Animal manure management systems (including centralised manure treatment plant where applicable); (c) Facilities which recover and flare/combust or use methane.

According to the methodology AMS-I.D, the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to.

Hence, the project boundary of the Project includes the physical and geographical sites of the livestock, the centralized animal manure management system, the biogas power plant and all the power plants connected into the NCPG.

Figure 3-1 describes the project boundary of the Project Activity.

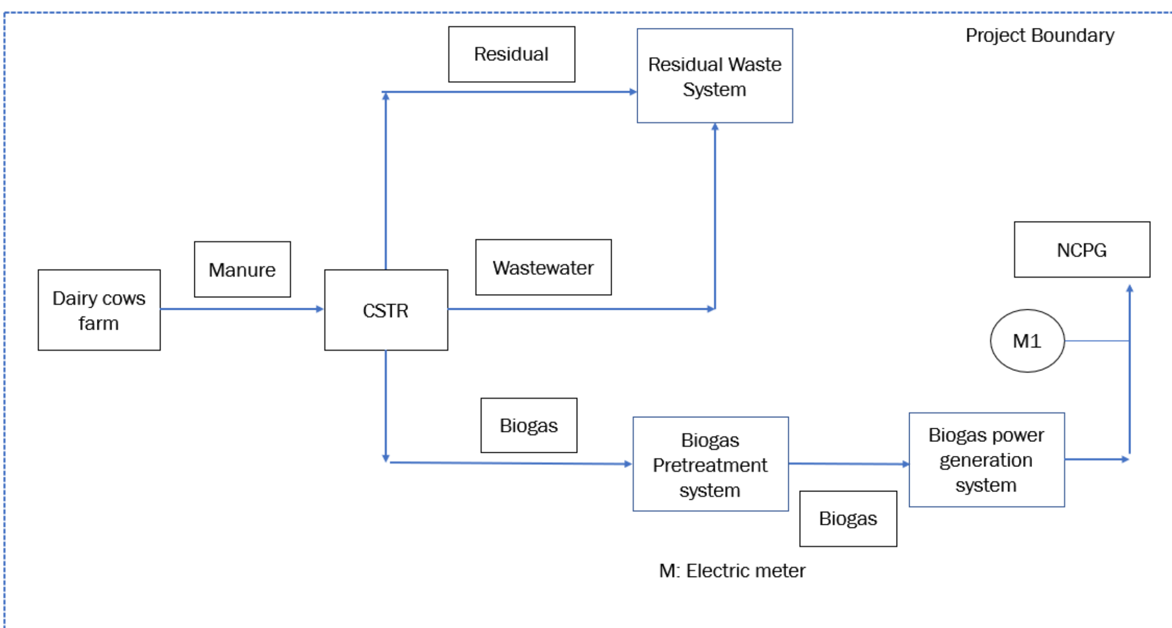


Table 3-4 Emission sources included in or excluded from the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Direct emissions from the manure treatment processes	CO_2	No	Excluded for simplification.
		CH_4	Yes	The major source of emissions in the baseline
		N_2O	No	Excluded for simplification.
	Emissions from electricity consumption /generation	CO_2	Yes	The major source of emissions in the baseline
		CH_4	No	Excluded for simplification.
		N_2O	No	Excluded for simplification.
Project	Physical leakage of biogas in the manure management systems	CO_2	No	Excluded for simplification.
		CH_4	Yes	Main emission source.
		N_2O	No	Excluded for simplification.
	Emissions from flaring or combustion of the gas stream	CO_2	No	Excluded for simplification.
		CH_4	No	Excluded for simplification. The enclosed flare is installed for emergency, the emissions are assumed to be very small. The project will not claim this part of emission reduction
		N_2O	No	Excluded for simplification.
	Emissions from use of fossil fuels or electricity	CO_2	Yes	The project consumes electricity during operation, so Emission from use of electricity is the main emission source. The project does not involve fossil fuel consumption, so the emission from use of fossil fuels is not included.
		CH_4	No	Excluded for simplification.
		N_2O	No	Excluded for simplification.
	Emissions from incremental transportation distances	CO_2	No	Excluded for simplification.
		CH_4	No	Excluded for simplification.
		N_2O	No	Excluded for simplification.
		CO_2	No	Excluded for simplification.

Source		Gas	Included?	Justification/Explanation
	Emissions from the storage of manure	CH_4	No	The storage time of the manure after removal from the animal barns, including transportation, is within 24 hours before being fed into the anaerobic digester, hence Emissions from the storage of manure is not accounted for.
		N_2O	No	Excluded for simplification.

3.4 Baseline Scenario

As per para. 17 of AMS-III.D, the baseline scenario is the situation where, in the absence of the project activity, animal manure is left to decay anaerobically within the project boundary and methane is emitted to the atmosphere.

As per para. 19 of AMS-I.D, the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Hence, the baseline scenario of the project is the animal manure waste was left to decay in anaerobic manure management system (open lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility; equivalent amount of electricity was supplied from the NCPG connected fossil fuel fired power plants.

3.5 Additionality

As per para. 15-16 of applied methodology AMS-III.D, project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site, that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the "Guidelines on the demonstration of additionality of small-scale project activities". This additionality condition also applies to Greenfield project activities. Furthermore, for project activities applying this methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component.

In line with AMS-III.D, the project recovers biogas to generate electricity with a total installed capacity of 2.33MW, which is lower than 5MW. The regulations relative to the project in China are identified as below.

a) Law of the People's Republic of China on Environment Protection;

b) Law of the People's Republic of China on the Prevention and Control of Solid Waste Pollution;

c) Law of the People's Republic of China on the Prevention and Control of Atmospheric Pollution;

d) Regulations on prevention and control of pollution from large scale livestock and poultry breeding;

e) Discharge standard of pollutants for livestock and poultry breeding (GB/T 18596);

f) Technical standard of pollution prevention for livestock and poultry breeding (HJ/T 81).

It has been identified that all the above laws and regulations in China **do not** require the collection and destruction of methane from livestock manure. In line with AMS-III.D, the project is deemed automatically additional.

3.6 Methodology Deviations

As plant specific fuel consumption and electricity generation data are not publicly available in China, the Guidance caused by DNV's request for deviation of Chinese project activities for the CDM baseline methodology AM0005 has been applied for calculating the build margin (BM) emission factor of this project activity. Moreover, the result of BM and OM are sourced from local government.

The deviation has no impact on conservativeness of the quantification of GHG emission reductions or removals.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

1. Baseline emissions from the manure treatment processes in year y (*$BE_{CH4,y}$*)

According to the methodology AMS-III.D, Baseline Emissions (BE_y) are calculated by using one of the following two options:

(a) Using the amount of the waste or raw material that would decay anaerobically in the absence of the project activity, with the most recent IPCC Tier 2 approach (please refer to the chapter 'Emissions from Livestock and Manure Management' under the volume 'Agriculture, Forestry and other Land use' of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories). For this calculation, information about the characteristics of the manure and of the management systems in the baseline is required. Manure characteristics include the amount of volatile solids (VS) produced by the livestock and the maximum amount of methane that can be potentially produced from that manure (B_o);

(b) Using the amount of manure that would decay anaerobically in the absence of the project activity based on direct measurement of the quantity of manure treated together with its specific volatile solids (SVS) content.

The project applies option (a) to calculate baseline emissions from the manure management (BE_y), and baseline emissions are determined as follows:

$$BE_y = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bl,j} \quad (1)$$

Where:

GWP_{CH4}	=	Global Warming Potential (GWP) of CH_4 applicable to the crediting period (tCO_2e/tCH_4)
D_{CH4}	=	CH_4 density (0. 67kg/m³ at room temperature (20ºC) and 1 atm pressure)
UF_b	=	Model correction factor to account for model uncertainties (0.94)
LT	=	Index for all types of livestock
j	=	Index for animal manure management system
MCF_j	=	Annual methane conversion factor (MCF) for the baseline animal manure management system j

$B_{0,LT}$	=	Maximum methane producing potential of the volatile solid generated for animal type LT (m^3CH_4/kg-dm)
$N_{LT,y}$	=	Annual average number of animals of type LT in year y (numbers)
$VS_{LT,y}$	=	Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)
$MS\%_{Bl,j}$	=	Fraction of manure handled in baseline animal manure management system j

a) The maximum methane-producing capacity of the manure ($B_{o,LT}$) varies by species and diet. The preferred method to obtain $B_{o,LT}$ measurement values is to use data from country-specific published sources, measured with a standardised method ($B_{o,LT}$ shall be based on total as-excreted VS). These values shall be compared to IPCC default values and any significant differences shall be explained. If country specific B_o values are not available, default values from tables 10 A-4 to 10 A-9 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10 can be used, provided that the project participants assess the suitability of those data to the specific situation of the treatment site.

b) VS are the organic material in livestock manure and consist of both biodegradable and non-biodegradable fractions. For the calculations the total VS excreted by each animal species is required.

   If country specific VS values are not available IPCC default values from 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 table 10 A-4 to 10 A-9 can be used provided that the project participants assess the suitability of those data to the specific situation of the treatment site particularly with reference to feed intake levels;

c) $B_{o,LT}$ values or VS values applicable to developed countries can be used provided the following four conditions are satisfied:

   i. The genetic source of the livestock originates from an Annex I Party;

   ii. The farm uses formulated feed rations (FFR) which are optimized for the various animal(s), stage of growth, category, weight gain/productivity and/or genetics;

   iii. The use of FFR can be validated (through on-farm record keeping, feed supplier, etc.);

   iv. The project specific animal weights are more similar to developed country IPCC default values.

d) Methane Conversion Factors (MCF) values are determined for a specific manure management system and represent the degree to which B_o is achieved. Where available country-specific MCF values that reflect the specific management systems used in particular countries or regions shall be used. Alternatively, the IPCC default values provided in table 10.17 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 can be used. The site annual average temperature is taken from official data at the nearest meteorological station, or from data available from historical on site observations.

e) The annual average number of animals ($N_{LT,y}$) is determined as follows:

$$N_{LT,y} = N_{da,y} \times \left(\frac{N_{p,y}}{365}\right) \quad (2)$$

Where:

$N_{da,y}$ = Number of days animal is alive in the farm in the year y (numbers)

$N_{p,y}$ = Number of animals produced annually of type LT for the year y (numbers)

2. Baseline emissions associated with electricity generation in year y (*$BE_{elec,y}$*)

As per AMS-I.D, baseline emissions associated with electricity generation in year y is calculated according to equation (3) as below:

$$BE_{elec,y} = EG_{PJ,y} \times EF_{grid,y} \quad (3)$$

Where:

$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh)
$EF_{grid,y}$	=	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO_2/MWh)

Calculation of combined margin CO_2 emission factor for grid (*$EF_{grid,y}$*)

According to "Tool to calculate the emission factor for an electricity system" (version 07.0). The following six steps are applied to determine OM, BM, and CM used for calculating the project baseline emissions:

- **Step 1:** Identify the relevant electricity systems;
- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
- **Step 3:** Select a method to determine the operating margin (OM);
- **Step 4:** Calculate the operating margin emission factor according to the selected method;
- **Step 5:** Calculate the build margin (BM) emission factor;
- **Step 6:** Calculate the combined margin (CM) emission factor.

As China DNA has published the calculation method for emission factor of grid, the published data and method have been applied for this project to calculate operating margin (OM) and build margin, as following steps:

Step 1: Identify the relevant electricity systems;

This project site is in Inner Mongolia Autonomous Region of China, which belongs to North China Power Grid (NCPG) according to the public delineation of DNA[9], so NCPG is identified as the relevant electric system.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);

For this project, Option I (only grid power plants are included in the calculation) is chosen.

Step 3: Select a method to determine the operating margin (OM);

Calculation of Operating Margin should be based on one of the four following methods according to the tool:

(a) Simple OM, or

(b) Simple adjusted OM, or

(c) Dispatch Data Analysis OM, or

(d) Average OM.

As the low-cost / must-run resources constituted less than 50% of total electricity generation of NCPG in recent five years (respectively 10.10%, 8.94%, 7,21%,6,39% and 5,92% in 2017, 2016, 2015, 2014 and 2013)[10], the Simple OM (a) method is selected and the following data vintage is chosen to calculate the emission factor:

Ex ante option: use a 3-year generation-weighted average, based on the most recent data available, without requirement to monitor and recalculate the emissions factor during the crediting period. And according to the tool, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

Step 4: Calculate the operating margin emission factor according to the selected method;

The Simple OM emission factor ($EF_{OM,y}$) is calculated as the generation-weighted average emissions per electricity unit (tCO_2e/MWh) of all generating sources serving in the system, excluding low-operating cost and must-run power plants. It may be calculated:

Option A: Based on the net electricity generation and a CO_2 emission factor of each power plant / unit, or

[9] http://cdm.ccchina.gov.cn/zyDetail.aspx?newsId=46143&TId=161

[10] China Electric Power Yearbook (2014, 2015, 2016, 2017, 2018)

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Option B can only be used if:

(a) The necessary data for Option A is not available; and

(b) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and

(c) Off-grid power plants are not included in the calculation.

In this project, all of the above conditions can be met, so Option B was chosen.

Under this option, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_i (FC_{i,y} \cdot NCV_{i,y} \cdot EF_{CO_2,i,y})}{EG_y} \quad (4)$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO_2 emission factor in year y (t CO_2/MWh)
$FC_{i,y}$	=	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO_2 emission factor of fuel type i in year y (t CO_2/GJ)
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in **Step 3.**

Based on the most recent three years (2015-2017) where the data are the latest and available at the time of this PD submission, the calculation result of $EF_{grid,OM,,y}$ is 0.9419 tCO_2e/MWh. The data is published by Ministry of Ecology and Environment of the People's Republic of China[11].

Step 5: Calculate the build margin (BM) emission factor;

As per Section 6.5 of TOOL07 (version 07.0), in terms of vintage of data, project participants can choose between one of the following two options:

(a) Option 1 - for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period;

(b) Option 2 - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

In line with China's Baseline emission factors of regional grids 2019 (BEF_{2019}) published by the Development and Reform Commission of China, Option 1 is chosen for the project; the BM emission factor is calculated ex ante based on the most recent information available on units already built for sample group m at the time of this project description submission.

The sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5\text{-}units}$) and determine their annual electricity generation ($AEG_{SET\text{-}5\text{-}units}$, in MWh);

(b) Determine the annual electricity generation of the proposed project electricity system, excluding power units registered as CDM project activities (AEG_{total}, in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the

[11] https://www.mee.gov.cn/ywgz/ydqhbh/wsqtkz/202012/t20201229_815386.shtml

generation of a unit, the generation of that unit is fully included in the calculation) (SET≥20%) and determine their annual electricity generation ($AEG_{SET≥20\%}$, in MWh);

(c) From $SET_{5-units}$ and SET≥20% select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. In this case ignore Steps (d), (e) and (f).

Otherwise:

(d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as CDM project activities, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample-CDM}$) the annual electricity generation ($AEG_{SET-sample-CDM}$, in MWh); If the annual electricity generation of that set comprises at least 20% of the annual electricity generation of the proposed project electricity system (i.e. $AEG_{SET-sample-CDM} \geq 0.2 \times AEG_{total}$), then use the sample group $SET_{sample-CDM}$ to calculate the build margin. Ignore steps (e) and (f).

Otherwise:

(e) Include in the sample group $SET_{sample-CDM}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sample-CDM->10yrs}$).

The BM emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which electricity generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m (EG_{m,y} \times EF_{EL,m,y})}{\sum_m EG_{m,y}} \quad (5)$$

where:

$EF_{grid,BM,y}$	=	Build margin emission factor of NCPG (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (t CO_2/MWh)
m	=	Power units included in the build margin
y	=	The most recent year for which the generation data is available

As it is difficult to obtain the detailed data on the power generation, fuel consumption and thermal efficiency of each newly built power unit from public documents, a deviation of TOOL07 (07.0) is adopted following the clarifications[12] given by the CDM EB concerning the BM emission factor calculation:

(1) The CDM EB suggested using the efficiency level of the best technology commercially available in the provincial/regional or national grid of China, as a conservative proxy, for each fuel type in estimating the fuel consumption to estimate the build margin.

(2) The EB agreed the use of capacity additions during last 1 ~ 3 years for estimating the build margin emission factor for grid electricity.

(3) The EB also agreed to use of weights estimated using installed capacity in place of annual electricity generation.

The newly built power plants in the past few years are bundled into "grouped new power plant" according to their construction year, their province and their fuel type. The annual net electricity generation in the year y of each "grouped new power plant" $EG_{m,y}$ is estimated according to their total capacity and the average utilization hours, as the following equation:

$$EG_{m,y} = CAP_m \times H_{m,y} \quad (6)$$

where:

$EG_{m,y}$	=	Annual net electricity generation the unit m in year y (MWh)
CAP_m	=	Installed capacity of the unit m (MW)
$H_{m,y}$	=	Utilization hour of the unit m in the year y (h), determined according to the average utilization hour of the same type of unit in the same province
y	=	The most recent year for which the generation data is available. For the calculation of BM in 2019, y = 2017

[12] "Request for clarification on use of approved methodology AM0005 for several projects in China", the EB's guidance on DNV deviation request. http://cdm.unfccc.int/UserManagement/FileStorage/AM_CLAR_QEJWJEF3CFBP1OZAK6V5YXPQKK7WYJ

m	=	grouped new power plant

Since the newly built power plants in the same province (A), in the same year (t) and using the same fuel type (k) are grouped into “a grouped new power plant”, CAP_m represents the total installed capacity of fuel type k power plants located in the provinces A and in the year t:

$$CAP_m = CAP_{A,t,k} \tag{7}$$

where:

CAP_m	=	Installed capacity of the unit m (MW), with m representing the specified combination of A, t, and k
$CAP_{A,t,k}$	=	Total installed capacity of fuel type k power plants located in the province A and in the year t
A	=	Provinces covered by the NCPG, namely, Beijing City, Tianjin City, Hebei Province, Shanxi Province, Shandong Province and Inner Mongolia Autonomous Region
t	=	Years related to the grouped new power plants, for the 2019 calculation, t represents 2017, 2016, 2015…. Until the aggregated electricity generation of the grouped new power plants reaches 20% of the total electricity generation of the NCPG
k	=	Fuel type of the grouped new power plants, including hydro, thermal (coal, gas, oil, waste incineration, other thermal), nuclear, wind, solar and others.

Figure 4 shows the procedure to determine the sample group of power units m.

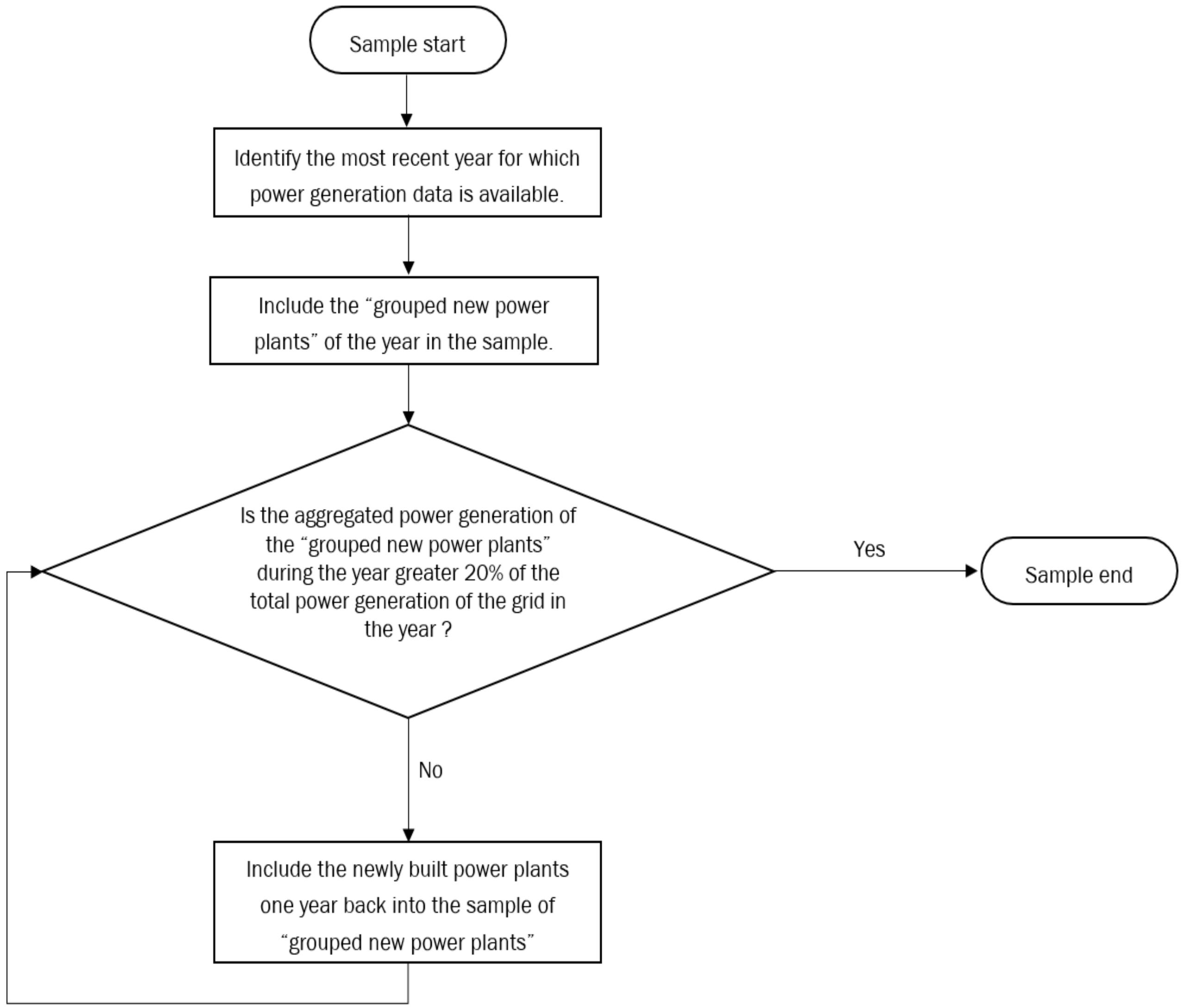


Figure 4 Procedure to determine the sample group of power units m used for the BM emission factor calculation

The emission factors of each fuel type $EF_{EL,m,y}$ are determined according to the Option A2 in the TOOL07, as the following equation:

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \quad (8)$$

where:

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (t CO_2/MWh)

$EF_{CO2,m,i,y}$	=	Average CO_2 emission factor of fuel type i used in power unit m in year y (t CO_2/GJ)
$\eta_{m,y}$	=	Average net energy conversion efficiency of power unit m in year y (ratio)
m	=	All power units serving the grid in year y except low-cost / must-run power units
3.6	=	Conversion factor (GJ/MWh)

Among the fuel types, the emission factors of hydro, nuclear, wind, solar, other thermal and others are 0. Concerning the emission factors of coal, gas, oil and waste incineration, Equation takes the following form due to conservativeness:

$$EF_{best,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{best,y}} \quad (9)$$

where:

$EF_{best,m,y}$	=	Emission factor of power unit m with the best technology commercially available in year y (t CO_2/MWh)
$\eta_{best,y}$	=	Power generation efficiency of the best technology commercially available in year y
m	=	Power units serving the grid with coal, gas, oil or waste incineration in year y

According to the latest and available data at the time of this PD submission, $EF_{grid,BM,y}$ is calculated to be 0.4819 tCO_2e/MWh. The data is published by Ministry of Ecology and Environment of the People’s Republic of China[13].

Step 6: Calculate the combined margin (CM) emission factor.

The calculation of the combined margin emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM.

The weighted average CM method (option A) should be used as the preferred option.

[13] https://www.mee.gov.cn/ywgz/ydqhbh/wsqtkz/202012/t20201229_815386.shtml

The simplified CM method (option b) can only be used if:

a) The project activity is located in: (i) a Least Developed Country (LDC); or in (ii) a country with less than 10 registered CDM projects at the starting date of validation; or (iii) a Small Island Developing States (SIDS); and

b) The data requirements for the application of step 5 above cannot be met.

This PD choose option A.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM} \quad 10)$$

Where:

$EF_{grid,OM,y}$	=	a) operating margin emission factor of NCPG (tCO_2e/MWh)
$EF_{grid,BM,y}$	=	b) build margin CO_2 emission factor of NCPG (tCO_2e/MWh)
w_{OM}	=	c) the weighting of operating margin emission factor (%)
w_{BM}	=	d) the weighting of build margin emission factor (%)

According to the tool, as a manure treatment project, w_{OM} = 0.5 and w_{BM} = 0.5 for the first crediting period .

$EF_{grid,CM,,y}$ = 0.9419 * 0.5 + 0.4819 * 0.5 = 0.7119 tCO_2e/MWh

4.2 Project Emissions

According to the methodology AMS-III.D and AMS-I.D, project emissions consist of:

(a) Physical leakage of biogas in the manure management systems which includes production, collection and transport of biogas to the point of flaring/combustion or gainful use ($PE_{PL,y}$);

(b) Emissions from flaring or combustion of the gas stream ($PE_{flare,y}$);

(c) CO_2 emissions from use of fossil fuels or electricity for the operation of all the installed facilities ($PE_{power,y}$);

(d) CO_2 emissions from incremental transportation distances;

(e) Emissions from the storage of manure before being fed into the anaerobic digester ($PE_{storage,y}$).

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y} \quad (11)$$

Where:

PE_y	=	Project emissions in year *y* (t CO_2e)
$PE_{PL,y}$	=	Emissions due to physical leakage of biogas in year *y* (t CO_2e)
$PE_{flare,y}$	=	Emissions from flaring or combustion of the biogas stream in the year *y* (t CO_2e)
$PE_{power,y}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year *y* (t CO_2e)
$PE_{transp,y}$	=	Emissions from incremental transportation in the year *y* (t CO_2e), as per relevant paragraph in AMS-III.AO
$PE_{storage,y}$	=	Emissions from the storage of manure (t CO_2e)

1. Emissions due to physical leakage of biogas in year y

Project emissions due to physical leakage of biogas from the animal manure management systems used to produce, collect and transport the biogas to the point of flaring or gainful use are estimated as:

$$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{i,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y} \quad (12)$$

Where:

GWP_{CH4}	=	Global Warming Potential (GWP) of CH_4 applicable to the crediting period (t CO_2e/t CH_4)
D_{CH4}	=	CH_4 density (0.00067 t/m^3 at room temperature (20 ºC) and 1 atm pressure)
LT	=	Index for all types of livestock
j	=	Index for animal manure management system
$N_{LT,y}$	=	Annual average number of animals of type LT in year y (numbers)
$VS_{LT,y}$	=	Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)
$MS\%_{i,y}$	=	Fraction of manure handled in system i in year y. If the project activity involves sequential manure management systems, the procedure specified in paragraph 18(e) of AMS-III.D shall be used to estimate the project emissions due to physical leakage of biogas in each stage
$B_{0,LT}$	=	Maximum methane producing potential of the volatile solid generated for animal type *LT* (m^3 CH_4/kg-VS), as per paragraph18 of AMS-III.D

2. Emissions from flaring or combustion of the biogas stream in the year y

In the case of flaring of the recovered biogas, project emissions are estimated using the procedures described in the methodological tool "Project emissions from flaring" (version 03.0).

Based on the FSR, all the methane generated by the project will be used as energy supply. In order to ensure no biogas is released under exigencies, an emergency flare system is installed at the project site, this emergency flare system is not used under normal operation. In addition, in case this emergency flare system operated, the emissions reductions during this period will be excluded for conservativeness, thus $PE_{flare,y}$ is excluded as well, so $PE_{flare,y}$ = 0 tCO_2e.

3. Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y

The project do not use fossil fuel. Thus, $PE_{FC,y}$ = 0 tCO_2e.

The electricity consumption of the project site can be met by the generated electricity. Thus, $PE_{EC,y}$ = 0 tCO_2e.

4. Emissions from incremental transportation in the year y

According to AMS-III.D, emission from transportation of animal manure between livestock farms and the project site is calculated using "option A: Monitoring fuel consumption" of the methodological tool "Project and leakage emissions from transportation of freight" (version 01.1.0). Based on the amount of fuel consumed by the vehicles, project emissions are determined using the latest version of the "Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion" (version 03.0). Parameter $PE_{FC,j,y}$ in the tool corresponds to the parameter $PE_{transp,y}$ of the project, which is calculated as follows:

$$PE_{transp,y} = \sum FC_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,y} \tag{16}$$

Where:

$PE_{transp,y}$	=	Emissions from incremental transportation in the year y (tCO_2)
$FC_{i,j,y}$	=	The quantity of fuel type i combusted in process j during the year y (mass or volume unit/yr)
$EF_{CO2,i,y}$	=	The weighted average CO_2 emission factor of fuel type i in year y (tCO_2/GJ)
$NCV_{i,y}$	=	The weighted average net calorific value of the fuel type i in year y (GJ/mass or volume unit)

The CSTR is installed within the farm, so there's no incremental transportation.

5. Emissions from the storage of manure

Project emissions on account of storage of manure before being fed into the anaerobic digester shall be accounted for if both condition (a) and condition (b) below are satisfied:

(a) The storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester;

(b) The dry matter content of the manure when removed from the animal barns is less than 20%.

The storage time of the manure after removal from the animal barns, including transportation, is 5~7 hours, within 24 hours before being fed into the anaerobic digester, hence emissions from the storage of manure is not accounted for, $PE_{storage,y}$ = 0 tCO_2e.

4.3 Leakage

As per "Project and leakage emissions from anaerobic digesters" (version 02.0), the leakage emissions associated with the anaerobic digester depend on how the digestate is managed. The leakage emissions include emissions associated with storage of digestate and composting of the digestate.

Digestate from the anaerobic digesters of the project will be sent to outside of project site for aerobic composting as soon as digestate is removed from the digesters and after composting, it will be returned to nearby farm as fertilizer. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve anaerobic composting of digestate. Therefore, leakage emissions of the project associated with the anaerobic digester is not accounted for.

4.4 Net GHG Emission Reductions and Removals

The emission reductions achieved by the project activity will be determined ex post through direct measurement of the amount of methane flared. It is likely that the project activity involves manure treatment steps with higher methane conversion factors (MCF) than the MCF for the manure treatment systems used in the baseline situation, therefore the emission reductions achieved by the project activity are limited to the ex post calculated baseline emissions minus the project emissions using the actual monitored data for the project activity (i.e. $N_{LT,y}$, $MS\%_{i,y}$, as well as $VS_{LT,y}$ in cases where adjusted values for animal weight are used). The emission reductions achieved in any year are the lowest value of the following:

$$\mathrm{ER}_{\mathrm{y,ex\ post}} = \min\left[\left(\mathrm{BE}_{\mathrm{y,ex\ post}} - \mathrm{PE}_{\mathrm{y,ex\ post}}\right), \left(\mathrm{MD}_{\mathrm{y}} - \mathrm{PE}_{\mathrm{power,y,ex\ post}}\right)\right] \tag{17}$$

Where:

$\mathrm{ER}_{\mathrm{y,ex\ post}}$	=	Emission reductions achieved by the project activity based on monitored values for year y (tCO_2e)
$\mathrm{BE}_{\mathrm{y,ex\ post}}$	=	Baseline emissions calculated using equation 1. For projects using option in paragraph 17(a) using ex post monitored values of $N_{LT,y}$ and if applicable $VS_{LT,y}$. For projects using option in paragraph 17(b), the ex post monitored values for $Q_{manure,j,LT,y}$ and $SVS_{j,LT,y}$ are used
$\mathrm{PE}_{\mathrm{y,ex\ post}}$	=	Project emissions calculated using equation 11 using ex post monitored values of $N_{LT,y}$, $MS\%_{i,y}$, and if applicable $VS_{LT,y}$
MD_{y}	=	Methane captured and destroyed or used gainfully by the project activity in year *y* (tCO_2e)
$\mathrm{PE}_{\mathrm{power,y,ex\ post}}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities based on monitored values in the year *y* (tCO_2e)

As per para.30 of AMS-III.D, if project activities utilize the recovered methane for power generation, may be calculated as follows, based on the amount of monitored electricity generation, without monitoring methane flow and concentration:

$$MD_y = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4} \quad (18)$$

Where:

EG_y	=	Total electricity generated from the recovered biogas in year *y* (MWh)
3600	=	Conversion factor (1 MWh = 3600 MJ)
NCV_{CH4}	=	NCV of methane (MJ/Nm^3) (use default value: 35.9 MJ/Nm^3)
EE_y	=	Energy conversion efficiency of the project equipment, which is determined by adopting one of the following criteria: Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and efficiency specification is for this fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation; Default efficiency of 40 %.

1. Calculation of baseline emissions:

Table 4-1: Ex-ante value of parameters to calculate BE_y

Parameter	Value	Data sources
GWP_{CH4}	28 tCO_{2e}/tCH_4	IPCC Fifth Assessment Report (AR5)
D_{CH4}	0.67 kg/m^3	AMS-III.D
UF_b	0.94	AMS-III.D
MCF_j	71%	Table 10.17 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
$B_{o,LT}$	Dairy cow:0.13 m^3CH_4/kg-VS	Table 10A-4 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
$VS_{default}$	Dairy cow: 2.8 kg/hd/day	Table 10A-4 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
$MS\%_{Bl,j}$	100%	FSR
nd_y	365 days	FSR
$N_{da,y}$	365 days	FSR

$N_{p,y}$	12,000	FSR

As per equations (1), (2), the result of the ex-ante calculated baseline emissions of methane from the manure treatment processes of the project is:

Table 4-2: Ex-ante Baseline Emissions BE_y

Index for all types of livestock (LT)	The annual average number of animals ($N_{LT,y}$)	Volatile solids production of livestock LT in year y ($VS_{LT,y}$) (kg-dm/animal/year)	Baseline Emissions (BE_y) tCO_2e
-	$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365})$	$VS_{LT,y} = VS_{default} \times nd_y$	$BE_y = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bl,j}$
Dairy cow	12,000	1,022	19,961
Total	-	-	19,961

As pre the FSR, the annual exported electricity to the NCPG of the project is expected to be 16,510 MWh, the combined margin CO_2 emission factor for the grid is 0.7119 tCO_2/MWh, hence the expected annual baseline emissions associated with electricity generation of the project is calculated as follows:

$BE_{elec,y} = EG_{PJ,y} \times EF_{grid,y}$ = 16,510 MWh× 0.7119 tCO_2/MWh = 11,753 tCO_2

$BE_y = BE_{CH4,y} + BE_{elec,y}$ = 19,961 tCO_2e + 11,753 tCO_2e = 31,714 tCO_2e

2. Calculation of project emissions

Table 4-3: Ex-ante value of parameters to calculate project emissions

Parameter	Value	Data sources
GWP_{CH4}	28 tCO_2e/tCH_4	IPCC Fifth Assessment Report (AR5)
D_{CH4} (ρ_{CH4})	0.67 kg/m^3	AMS-III.D
$B_{o,LT}$	Dairy cow:0.13 m^3CH_4/kg-VS	Table 10A-4 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
$VS_{default}$	Dairy cow: 2.80 kg/hd/day	Table 10A-4 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
$MS\%_{i,y}$	100%	FSR
$N_{da,y}$	365 days	FSR

$N_{p,y}$	12,000	FSR
$Q_{biogas,d}$	8,760,000 m³/year	FSR
$f_{CH4,default}$	0.6	"Tool to Project and leakage emissions from anaerobic digesters"

2.1 Emissions due to physical leakage of biogas in year y

As described in 4.2, project emissions from physical leakage of biogas are calculated as per equations (2), (12).

Table 4-4: Ex-ante calculate Physical leakage of biogas ($PE_{PL,y}$)

Index for all types of livestock (LT)	The annual average number of animals ($N_{LT,y}$)	Volatile solids production of livestock LT in year y ($VS_{LT,y}$) (kg-dm/animal/year)	Physical leakage of biogas ($PE_{PL,y}$) tCO_2e
-	$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365})$	$VS_{LT,y} = VS_{default} \times nd_y$	$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{j,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y}$
Dairy cow	12,000	1,022	2,991
Total	-	-	2,991

So, the ex-ante calculated result of the project's Physical leakage of biogas is 2,991 tCO_2e.

2.2 Emissions from flaring or combustion of the biogas stream in the year y

As described above, $PE_{flare,y}$ is excluded, hence $PE_{flare,y}$ = 0 for the project.

2.3 Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y

As described in 4.2, the project does not use any fossil fuel, so $PE_{FC,y}$ is not included in the project emission.

As described in 4.2, the electricity consumption of the project site can be met by the generated electricity. Thus, $PE_{EC,y}$ = 0.

2.4 Emissions from incremental transportation in the year y

As described above, $PE_{transp,y}$ is excluded, hence $PE_{transp,y}$ = 0 for the project.

2.5 Emissions from the storage of manure

As described above, $PE_{storage,y}$ is excluded, hence emissions from the storage of manure are not accounted for, $PE_{storage,y}$ = 0 tCO_2e.

So, the ex-ante estimated project emission shall calculate as per equation (10):

$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y} = PE_{PL,y} = 2{,}991 tCO_2e$

3. Calculation of leakage

As per “Project and leakage emissions from anaerobic digesters” (version 02.0), the leakage emissions associated with the anaerobic digester depend on how the digestate is managed. The leakage emissions include emissions associated with storage of digestate and composting of the digestate.

Digestate from the anaerobic digesters of the project will be used to produce fertilizer as soon as digestate is removed from the digesters. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve composting of digestate. Therefore, leakage emissions of the project associated with the anaerobic digester is not accounted for.

Thus, $LE_y = 0$.

4. Calculation of emission reductions (Ex ante calculation)

Estimated annual emission reductions of the project can be calculated as following equation (19):

$$ER_y = BE_y - PE_y - LE_y = BE_y - PE_y \quad (19)$$

Year	Estimated baseline emissions or removals (tCO2e)	Estimated project emissions or removals (tCO2e)	Estimated leakage emissions (tCO2e)	Estimated net GHG emission reductions or removals (tCO2e)
26-August-2022~31-December-2022	11,122	1,049	0	10,073
01-January-2023~31-December -2023	31,714	2,991	0	28,723
01-January -2024~31-December -2024	31,714	2,991	0	28,723
01-January -2025~31-December -2025	31,714	2,991	0	28,723
01-January -2026~31-December -2026	31,714	2,991	0	28,723
01-January -2027~31-December -2027	31,714	2,991	0	28,723
01-January -2028~31-December -2028	31,714	2,991	0	28,723
01-January -2029~25-August -2029	20,592	1,942	0	18,650
Total	**221,998**	**20,937**	**0**	**201,061**

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	GWP_{CH4}
Data unit	tCO_2e/tCH_4
Description	Global Warming Potential (GWP) of CH_4 applicable to the crediting period
Source of data	IPCC
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value of 28 from IPCC Fifth Assessment Report (AR5). Shall be updated according to any future revision to VCS standard by VERRA.
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	-

Data / Parameter	D_{CH4}
Data unit	kg/m^3
Description	CH_4 density
Source of data	AMS-III.D
Value applied	0.67 (at 20 °C and 1 atm pressure)
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	-

Data / Parameter	UF_b
Data unit	-

Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.D
Value applied	0.94
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	-

Data / Parameter	MCF_j
Data unit	-
Description	Annual methane conversion factor (MCF) for the baseline animal manure management system j
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	71%
Justification of choice of data or description of measurement methods and procedures applied	No country or regional specific value is available. Default value from table 10.17 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 is applied. The annual average temperature of baseline site where anaerobic manure treatment facility is located is around 13°C[14], the corresponding annual methane conversion factor (MCF) is 71%.
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	-

Data / Parameter	$B_{0,LT}$
Data unit	$m^3\ CH_4$/kg-VS
Description	Maximum methane producing potential of the volatile solid generated for animal type LT

[14] https://www.hd.gov.cn/zjhd/hdgk/zrdl/

Source of data	Table 10A-4 of Volume 4 Chapter 10 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	Animal type: Dairy cow; $B_{0,LT}$: 0.13
Justification of choice of data or description of measurement methods and procedures applied	No country or regional specific value is available. Default values from tables 10 A-4 to 10 A-9 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10 are applied. The project adopts Asian data.
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	-

Data / Parameter	$MS\%_{BL,i}$
Data unit	-
Description	Fraction of manure handled in baseline animal manure management system j
Source of data	FSR
Value applied	100%
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	-

Data / Parameter	$N_{da,y}$
Data unit	Day
Description	Number of days animal is alive in the farm in the year y
Source of data	Project proponents

Value applied	365
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	To calculate the annual average number of animals ($N_{LT,y}$)
Comments	-

Data / Parameter	$EF_{grid,OM,y}$
Data unit	t CO_2/MWh
Description	Operating margin CO_2 emission factor in year y
Source of data	"2019 Baseline Emission Factors for Regional Power Grids in China" published by Ministry of Ecology and Environment of China
Value applied	0.9419
Justification of choice of data or description of measurement methods and procedures applied	Official and authoritative statistic data.
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	According to section 4.1, the ex ante option is selected to calculate $EF_{grid, OM,y}$ and fixed in the crediting period.

Data / Parameter	$EF_{grid,BM,y}$
Data unit	t CO_2/MWh
Description	Build margin CO_2 emission factor in year y
Source of data	"2019 Baseline Emission Factors for Regional Power Grids in China" published by Ministry of Ecology and Environment of China
Value applied	0.4819
Justification of choice of data or description of measurement methods and procedures applied	Official and authoritative statistic data.

Purpose of Data	Calculation of baseline emissions and project emissions
Comments	According to section 4.1, the ex ante option is selected to calculate $EF_{grid,BM,y}$ and fixed in the crediting period.

Data / Parameter	$EF_{grid,CM,y}$ ($EF_{grid,y}$, $EF_{EL,j,y}$)
Data unit	t CO_2/MWh
Description	Combined margin emission factor for the grid in year y
Source of data	Calculated based on “Tool to calculate the emission factor for an electricity system” (version 07.0).
Value applied	0.7119
Justification of choice of data or description of measurement methods and procedures applied	$EF_{grid,CM,y}$ is calculated based on $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ as per the latest version of “Tool to calculate the emission factor for an electricity system”
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	According to section 4.1, the ex ante option is selected to calculate $EF_{grid, OM,y}$ and $EF_{grid,BM,y}$ and fixed in the crediting period, thus the $EF_{grid,CM,y}$ is fixed during the crediting period.

Data / Parameter	NCV_{CH4}
Data unit	MJ/Nm^3
Description	NCV of methane
Source of data	AMS-III.D
Value applied	35.9
Justification of choice of data or description of measurement methods and procedures applied	Default value from methodology is applied
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	$w_{CH4,y}$
Data unit	%
Description	Methane content in biogas in the year y
Source of data	AMS-III.D default value
Value applied	60%
Justification of choice of data or description of measurement methods and procedures applied	Default value of 60% is applied as per AMS-III.D
Purpose of Data	To calculate the emission reductions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$MS\%_{i,y}$
Data unit	-
Description	Fraction of manure handled in system i in year y
Source of data	Project proponents
Description of measurement methods and procedures to be applied	all manure would be handled in this project, 100% is applied for $MS\%_{i,y}$ in ex-ante estimation. This parameter will be monitored in verification period.
Frequency of monitoring/recording	-
Value applied	100%
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions and project emissions
Calculation method	-
Comments	-

Data / Parameter	$N_{p,y}$
Data unit	Number
Description	Number of animals produced annually of type LT for the year y
Source of data	Project proponents
Description of measurement methods and procedures to be applied	The number of Dairy cows produced in the farm will be recorded manually by the responsible staff.
Frequency of monitoring/recording	Annually, based on monthly records
Value applied	12,000
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	To calculate the annual average number of animals ($N_{LT,y}$)
Calculation method	-
Comments	-

Data / Parameter	nd_y
Data unit	number
Description	The number of days the treatment plant was operational in year y.
Source of data	Project proponents
Description of measurement methods and procedures to be applied	365 days used for ex-ante estimation. The actual number of days the treatment plant was operationally used in the monitoring periods will be monitored and recorded by staff.
Frequency of monitoring/recording	Annually, based on daily records and monthly aggregation
Value applied	365
Monitoring equipment	-
QA/QC procedures to be applied	-

Purpose of data	To estimate the annual volatile solid excretions for livestock LT entering all animal waste management systems on a dry matter weight basis ($VS_{LT,y}$).
Calculation method	-
Comments	-

Data / Parameter	$VS_{default}$
Data unit	kg/hd/day
Description	Default value for the volatile solid excretion rate per day on a dry-matter basis for a defined livestock population
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories
Description of measurement methods and procedures to be applied	According to animals' genetic region and type, Default values from tables 10 A-8 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10 are applied. The genetic source for all types of Dairy cows is from Asia, the average weight of Dairy cow is 360kg, which is slightly higher than the default value (350kg) of Region "Asia" from 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 chapter 10 table 10 A-8. Thus, VS value from IPCC default value can be used, which is 2.80kg/hd/day and it is conservative. The animal purchase contract, Feed Purchase Contract and on-farm record keeping will be monitored in verification period.
Frequency of monitoring/recording	Annually
Value applied	Animal type: Dairy cows; $VS_{default}$: 2.80
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions and project emissions
Calculation method	-
Comments	-

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh
Description	Quantity of electricity generation supplied by the project to the grid in year y
Source of data	Electricity meter
Description of measurement methods and procedures to be applied	Measured by electricity meter continuously and monthly recorded
Frequency of monitoring/recording	Continuous monitoring and at least monthly recording
Value applied	16,510 (Ex-ante estimation)
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Sales receipts are to be used for crosscheck. The electricity meter(s) will be periodically checked and maintained. Calibration will be carried out once a year to ensure normal operation. The accuracy of the meter is not less than 0.5s.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	EG_y
Data unit	MWh
Description	Total electricity generated from the recovered biogas in year y
Source of data	Electricity meter
Description of measurement methods and procedures to be applied	Measured by electricity meter installed at the outlet of biogas generators
Frequency of monitoring/recording	Continuously monitored and monthly recorded
Value applied	16,510 (Ex-ante estimation)

Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Sales receipts are to be used for crosscheck. The electricity meter(s) will be periodically checked and maintained. Calibration will be carried out once a year to ensure normal operation. The accuracy of the meter is not less than 0.5s.
Purpose of data	Calculation of emission reductions
Calculation method	-
Comments	-

Data / Parameter	EE_y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Source of data	Specification provided by the equipment manufacture
Description of measurement methods and procedures to be applied	As per paragraph 30 of the methodology. Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and the efficiency specification is for this fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
Frequency of monitoring/recording	-
Value applied	Monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of emission reductions
Calculation method	-
Comments	-

5.3 Monitoring Plan

The monitoring plan presented in this PD assures that real, measurable, long-term GHG emission reductions can be monitored, recorded and reported. It is a crucial procedure to identify the final

VCUs of the project. This monitoring plan will be implemented by the project owner during the project operation. The details of the monitoring plan are specified as follows:

(A) Monitoring structure

The Project owner organizes a specific VCS team in project development department to be responsible for data collection, supervision and witness the whole process of data measuring and recording. A VCS manager is appointed to take full responsibility for the overall monitoring of the project. The monitoring and measurement is to be carried out by designated monitoring officers. In addition, the Project developer appoints internal verifiers who is responsible for internal check of the measurement, collection of relevant receipts and invoices, and the calculation of the emission reductions. A monitoring and management manual of the project that identifies detailed duties and responsibilities of the relevant parties is developed and served as the basis of the project monitoring. Figure 5-1 shows the operation and management structure of the Project.

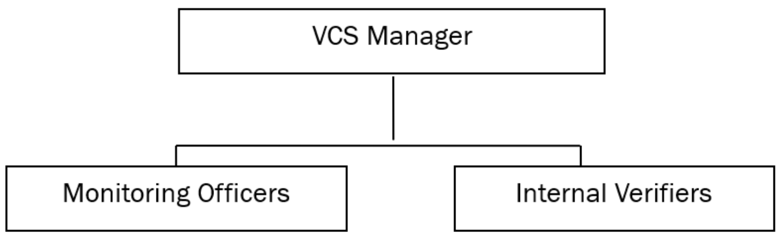


Figure 5-1 Operation and management structure of the project

(B) Data and parameters to be monitored

Data and parameters to be monitored are listed below. Figure 5-2 shows the positions of the monitoring instruments and Table 5-1 lists the corresponding parameters monitored:

Table 5-1 Data and parameters to be monitored

Position	Parameter Monitored	Description
M1	$EG_{PJ,y}/EG_y$	Quantity of electricity generation supplied by the project to the grid in year y continuously measured by electricity meter and at least monthly recorded

(C) Data collection

Monitoring officers are responsible for data collection. Designated teams will read and collect the monitored data regularly. The computer system will automatically monitor and record relevant meter data. Automatic records will serve as the main data source for emission reductions calculation. All data files, relevant receipts will be collected by a designated monitoring officer, who will prepare backup in time and archive all documents properly.

(D) Quality assurance

All metering equipment for monitoring will be chosen in accordance with VCS requirements and will be calibrated regularly for accuracy by qualified party according to the national regulations. To assist in future verifications, the Project owner will preserve the calibration records, along with the data files of project monitoring.

Error check routines will be established on site and at the point of data storage to detect data measuring/transmission failures as well as malfunctions. In the case of malfunction of the meters, the meter supplier will provide technical support to engage the problem promptly and emission reductions during the corresponding period will be calculated conservatively.

The installation of the electricity metering equipment will fulfill the requirements of "*DL/T448–2016 Technical Administrative Code of Electric Energy Metering*". The accuracy should not be less than 0.5. All the meters will be checked and maintained periodically.

(E) Data file management

All monitoring data will be electronically filed by the end of each month and the electronic data files will be archived in both disk copy and printed hard copy. Other documents in paper e.g. forms and environment assessment reports will be preserved as well. All data collected as part of monitoring will be archived electronically and be kept at least for 2 years after the end of the crediting period. The Project owner will provide original records and documents if necessary.